POPS Worldwide: Offering quality streaming content to a rapidly growing audience

Tell us your challenge. We're here to help.

Contact us (/contact)

About POPS

About POPS Worldwide

Founded in 2007, POPS Worldwide is a leading digital entertainment company in Southeast Asia. Providing quality streaming content through its two applications and major global online entertainment platforms, the company has more than 400 million subscribers and 4.3 billion average monthly views. Based in Vietnam, POPS Worldwide started its expansion to Thailand in 2015 and Indonesia in late 2020.

Industries: Media & Entertainment

To support the ra
growth of its con
streaming applic
its market expan
POPS Worldwide
Google Cloud for
scalable, easily m
cloud infrastruct

requiring lower op
costs.

Location: Vietnam

Storage (https://cloud.google.com/storage/),

Google Cloud (https://cloud.google.com/)

(https://cloud.google.com/kubernetes-engine/),

BigQuery (https://cloud.google.com/bigquery/)

Cloud Storage (https://cloud.google.com/storage/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Pub/Sub (https://cloud.google.com/pubsub/)

Recommendations AI (https://cloud.google.com/recommendations)

Contact us (/contact)

Google Cloud results

- Handles 40x app traffic growth with highly scalable cloud infrastructure
- Supports quick expansion to two foreign markets in less than one year with cross-border network
- Helps save 30% of operational costs per content minute with cost-effective use of computing

Zero infras syste two y

resources on GKE
- Enables a small DevOps team of two to cover infrastructure maintenance, freeing more workforce for innovation

The online media streaming market in Southeast Asia [is expanding at an unprecedented speed](https://fivestones.net/en/blog/digital-marketing/explosion-of-streaming-services-asia-pacific/) (https://fivestones.net/en/blog/digital-marketing/explosion-of-streaming-services-asia-pacific/). A growing tech-savvy population coupled with the arrival of mobile devices with high display resolution and improved internet connection has resulted in online streaming platforms becoming a primary form of entertainment. The trend has been further accelerated by COVID-19, as a recent survey (https://www.businesswire.com/news/home/20201207006003/en/57-Percent-of-Southeast-Asian-Viewers-are-Now-Streaming-More-OTT-Video-Content-Because-of-COVID-19-According-to-New-Research-from-The-Trade-Desk) shows that 57% of Southeast Asian online video viewers are now streaming more content because of the pandemic.

Among the wide range of online streaming providers, POPS Worldwide (https://popsww.com/en/) stands out as one of the leading digital entertainment companies in Vietnam with its carefully selected, high-quality content tailored for all age groups. Founded in 2007, the company started by distributing music and videos through major global online entertainment platforms like YouTube and launched its two streaming applications, POPS Kids in 2017, and POPS in 2019. Boasting a diverse library of digital content across entertainment and edutainment, POPS Worldwide currently has more than 400 million subscribers and 4.3 billion average monthly views, with the viewership doubling every year.

"We are especially impressed with the support we have been receiving from Google Cloud. Our questions are always quickly answered, and with a lot of training materials available, we

could complete
our move to
Google Cloud
faster than
expected."

—Martin Papy, CTO, POPS Worldwide

In early 2019, when POPS Worldwide saw a continuous increase in its app traffic, it began to realize that the private cloud where its apps were deployed at that time fell short in scalability, stability, and cost-effectiveness. Its existing cloud had little automation and required daily maintenance. At the same time, the company was preparing to enter foreign markets and found that relying on a locally hosted cloud infrastructure would greatly delay its expansion plans.

To solve these two issues, POPS Worldwide decided to migrate its two streaming applications to a public cloud environment and eventually turned to Google Cloud (https://cloud.google.com/) because of its familiarity with Google products, such as having decade-long content distribution activities on YouTube, and using Google Workspace (https://workspace.google.com/) as its default internal communications platform. The support provided by the Google Cloud team further convinced POPS Worldwide that Google Cloud was the right choice. “Moving to Google Cloud for us was a natural choice, since we already use BigQuery (/bigquery) to store data received from our YouTube channels. Furthermore, we have already established good access management for our employees on Google Workspace," recalls Martin Papy, CTO at POPS Worldwide.

It took the team fewer than four months to complete the migration process. Martin says, "We are especially impressed with the support we have been receiving from Google Cloud. Our questions are always quickly answered, and with a lot of training materials available, we could complete our move to Google Cloud faster than expected."

Supporting app traffic surge and market expansion with a highly scalable cloud infrastructure

POPS Worldwide migrated its streaming applications to the virtual machines (VMs) on Compute Engine (/compute). Cloud Storage (/storage) helps store all the content on its apps and connect with all its content distribution networks. POPS

Worldwide also leverages BigQuery to process data collected on its apps and conduct various analyses ranging from financial to app traffic reports. On top of that, Pub/Sub (/pubsub) is used as a buffer for data collection to ensure smooth data transfer when there is a spike in volume.

One of the biggest advantages that the POPS Worldwide team noticed after moving to Google Cloud is the enhanced scalability. Over the past year, as COVID-19 prompted people to stream more content online, the company's app traffic experienced a huge surge. The company has also been able to launch marketing campaigns that result in a 100x increase in traffic volume. Martin says that the team would have not been able to handle all these traffic increases if the apps were still running on a private cloud.

“Since we moved to Google Cloud in May 2019, our traffic volume has grown around 40 times. We simply couldn’t have achieved this growth if we stayed on our previous infrastructure, because the cost would have been astronomic,” explains Martin. “Before, we ran only eight VMs on the private cloud. Today, we are running more than 100 VMs on Google Cloud and can launch new VMs at any time with just a few clicks. We would have never been able to reach this scale in the past.”

In the past, POPS Worldwide's system would face system errors, which would sometimes disrupt user experience. There has been a significant improvement to the app performance after the migration.

“With Google Cloud, we have never encountered infrastructure or connectivity issues in the past two years. This has greatly improved the user experience of our apps and helped us retain and attract more audience,” Martin says.

At the same time, the high scalability and well-connected cross-border network of Google Cloud have assisted POPS Worldwide to complete foreign market expansion in a short period of time, as it is able to roll out its apps in a new country using the same infrastructure. The company successfully launched its apps in Thailand in late 2020, according to its plan, and is preparing to enter the Indonesian market by 2022.

“We wanted to use the resources on Google Cloud more efficiently. By providing more flexible VM specifications, GKE allows us to have a more cost-effective resource allocation for each service that we are using and helps us realize cost optimization."

—Martin Papy, CTO, POPS Worldwide

Reducing operational costs with more cost-effective use of resources on GKE

Now, POPS Worldwide is seeking ways to optimize its costs for further growth. As a part of this effort, the company moved its streaming apps from Compute Engine to the VMs on Google Kubernetes Engine (/kubernetes-engine) (GKE), where it can fine-tune the use of computing resources according to its needs. The ability to use VM CPUs and memory more efficiently on GKE immediately helped the team save 30% of backend operational cost.

"We wanted to use the resources on Google Cloud more efficiently. By providing more flexible VM specifications, GKE allows us to have a more cost-effective resource allocation for each service that we are using and helps us realize cost optimization," notes Martin. "With Google Cloud, there are always tools available for optimizing our services and operations."

Furthermore, the automation feature of GKE enables POPS Worldwide to manage the cloud infrastructure with only two full-time DevOps engineers, giving the team more resources to develop new features and create quality streaming content.

Building more powerful content recommendation engine with Recommendations AI

To offer its app users more personalized content recommendations and further increase its viewership, POPS Worldwide has been inventing a recommendation engine powered by artificial intelligence (AI) since 2020. After building the first two models using its own and open source AI technologies, the company started to take advantage of the more intuitive and powerful features of Recommendations AI (/recommendations) to develop its third-generation recommendation model.

“Using Recommendation AI to build a machine learning (ML) model does not require too much expertise in AI, which allows us to proceed with the project with a very small team without asking for external help,” says Martin.

He adds that compared with other AI tools, the team is also able to add more parameters on Recommendation AI to train the ML model and deliver more accurate output. The fact that Recommendation AI can easily connect data with Cloud Storage and BigQuery has also accelerated the development process.

“Google Cloud has equipped us with sufficient tools to realize rapid growth and focus on improving our streaming content, and we look forward to leveraging all the solutions that can bring us more scalability and cost-effectiveness to achieve more."

—Martin Papy, CTO, POPS Worldwide

Further harnessing cloud scalability and power of data for greater business results

Moving forward, POPS Worldwide plans to harness the high scalability of Google Cloud products for further cost optimization and make full use of BigQuery to unlock the power of data for customer recommendations and improve the customer experience. It also plans to use BigQuery ML (/bigquery-ml/docs) to predict viewing trends on its streaming apps. POPS Worldwide will also be moving its legacy code currently running on its local database to Cloud Bigtable (/bigtable), which will provide higher computing speed and scalability in a cost-effective way.

"The online streaming market is growing faster than ever in Southeast Asia, and we are constantly improving our services and operations to stay competitive," says Martin. "Google Cloud has equipped us with sufficient tools to realize rapid growth and focus on improving our streaming content, and we look forward to leveraging all the solutions that can bring us more scalability and cost-effectiveness to achieve more.”